UCH409: PROCESS ENGINEERING SIMULATION SOFTWARES

L	T	P	Cr
2	0	2	3.0

Course Objective:
To impart the basic knowledge and handling of the software used in simulation, design, and optimization ofa chemical process plant. It includes the software related to the material and energy balancing, equipment sizing, and costing calculation and core activities in process engineering.

Introduction: The role of process Modeling, simulation, and optimization in the chemical and process industries. Idea about general software tool such as Excel, MATLAB, etc. for chemical engineers.

Excel for Chemical Engineering: Basic Spreadsheet Skills, Configuring Excel for engineering calculations,Efficient spreadsheet manipulations, Formulas, cell addressing and range names, creating engineering graphs, dealing with engineering formulas and units, debugging spreadsheet calculations, Process Calculations, Flowsheet calculations with recycle, targeting calculations, etc. Case studies related to fluid mechanics, heat transfer and unit operation problems using excel such as piping design, pump design, coolingtower, etc.

MATLAB and Simulink for Chemical Engineering: Basic commands and functions, 2D plotting functions, plotting with multiple horizontal or vertical **axes**, plots of implicit functions, plotting 3D curves and surfaces, fundamental matrix operations, Interpolation and data fitting, Introduction to Simulink, Simulink parameters and solvers, regression analysis using Simulink, construction of block diagrams.

CFD and Multiphysics Modeling and simulation: Introduction to CFD, and CFD software introduction toANSYS Fluent and COMSOL Multiphysics, problem definition, geometry creation mesh generation and mash optimization, data input, physics selection, simulation, post-processing of the results and analysis. Case studies related to fluid mechanics, heat transfer, mass transfer, reactor design and dynamics such as pipe flow, heat exchanger, PFR, CSTR, etc.

Course learning outcomes (CLOs):
Upon completion of this course, the student will able to:
1. solve the chemical engineering problems in Excel.
2. simulate the chemical engineering problems in MATLAB.
3. perform the CFD simulation in ANSYS Fluent and COMSOL Multiphysics.

Text Books:
1. *Martín M.M., Introduction to Software for Chemical Engineers, CRC press, 2nd ed., 2020.*
2. *Yeo Y.K., Chemical Engineering Computation with MATLAB, CRC press, 2020.*
3. *Al-Malah K.I.M., ASPEN PLUS® Chemical Engineering Applications, Wiley, 2017.*
4. *Chaurasia A.S., Computational Fluid Dynamics and COSMOL Multiphysics - A Step-By-Step Approach for Chemical Engineers, 1st ed. CRC press, 2022.*

Reference Books:

1. *MATLAB manual.*
2. *ASPEN PLUS manual.*
3. *ANSYS Fluent manual.*
4. *COMSOL Multiphysics manual.*

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	45
3.	Sessional (may include Quizzes/Assignments/Lab evaluations)	30